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Reversing Pressure Decline in a Mature Giant Carbonate Reservoir Under High Production Demand Through Adaptive Field Development Strategy and Smart Surveillance

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Abstract

Carbonate reservoirs often exhibit complex pore structure, strong heterogeneity, and unpredictable fluid-flow behaviors - factors that can undermine water injection effectiveness and jeopardize reservoir energy sustainability. In these systems, achieving effective waterflood conformance and maintaining pressure support pose technical challenges that vary with geological architecture and development history. Aiming to overcome these constraints, this paper describes the adaptive field development plan implemented in a mature giant carbonate reservoir (A) of 21x5 km² within Shuaiba formation in Abu Dhabi, with the objective of restoring reservoir pressure through improved voidage replacement, optimized injection distribution, and the application of smart surveillance workflows to sustain long-term production.

The strategy combined subsurface and surface reviews in an integrated workflow to ensure water was available and injected where required to arrest pressure decline and meet production requirements during periods of high demand. The subsurface review, based on the revised Field development plan, incorporated a new sectorization and flow-unit classification derived from the latest geological model update. This framework improved the understanding of reservoir heterogeneity. Water injection requirements were reassessed, leading to a redistribution of support from peripheral to new roll-up scheme to improve areal and vertical conformance, along with stimulation of underperforming injectors. To ensure timely pressure recovery, the line-drive configuration, although part of the longterm development plan, was implemented earlier to mitigate insufficient support in the central area. In parallel, the surface network was evaluated to identify capacity constraints and optimize cluster configurations to close water distribution gaps. These actions were supported by enhanced surveillance, integrating production data, permanent down-hole gauge data and static surveys for non-instrumented wells, all within validated datasets consolidated into Spotfire® dashboards. This approach functioned as smart surveillance, providing rapid and efficient analysis that facilitated fast decision-making across wells, sectors and entire reservoir A.

The decline in reservoir pressure was successfully arrested, with clear recovery observed in several wells - particularly in the flanks, while central areas reported stabilization with gradual improvement. These outcomes were achieved despite operating at high-rate conditions that exceeded the planned targets for some periods of time, which delayed full restoration but provided a robust validation of the revised development plan. Injection capacity was ramped up by about 155% with debottlenecking efforts in surface water capacity, the addition of new roll-up and line-drive wells, and stimulation campaign, enabling more effective distribution of injection water. Monitoring capabilities also advanced with a reduction of 50% of required shut-in period for static pressure, allowing more frequent pressure surveys with a reduced impact on production. The distinctive aspect of this work is the integration of geological updates, injection management, automated surveillance analysis and cross-disciplinary data can support adaptive field development decisions and sustain production under high-demand conditions.

Introduction

Carbonate reservoirs in the Middle East hold some of the world's largest oil accumulations and present persistent challenges for pressure maintenance and displacement efficiency. Their complex rock fabric, with vugs, fractures and strong heterogeneity mainly in permeability, creates bypassed oil during water injection, leading to early water breakthrough (Xu et al., 2020).

Many mature carbonate fields in the region are now experiencing pressure decline despite decades of water injection. Limitations in surface facilities, poor injection distribution and increasing offtake often accelerate this trend, reducing productivity and long-term recovery.

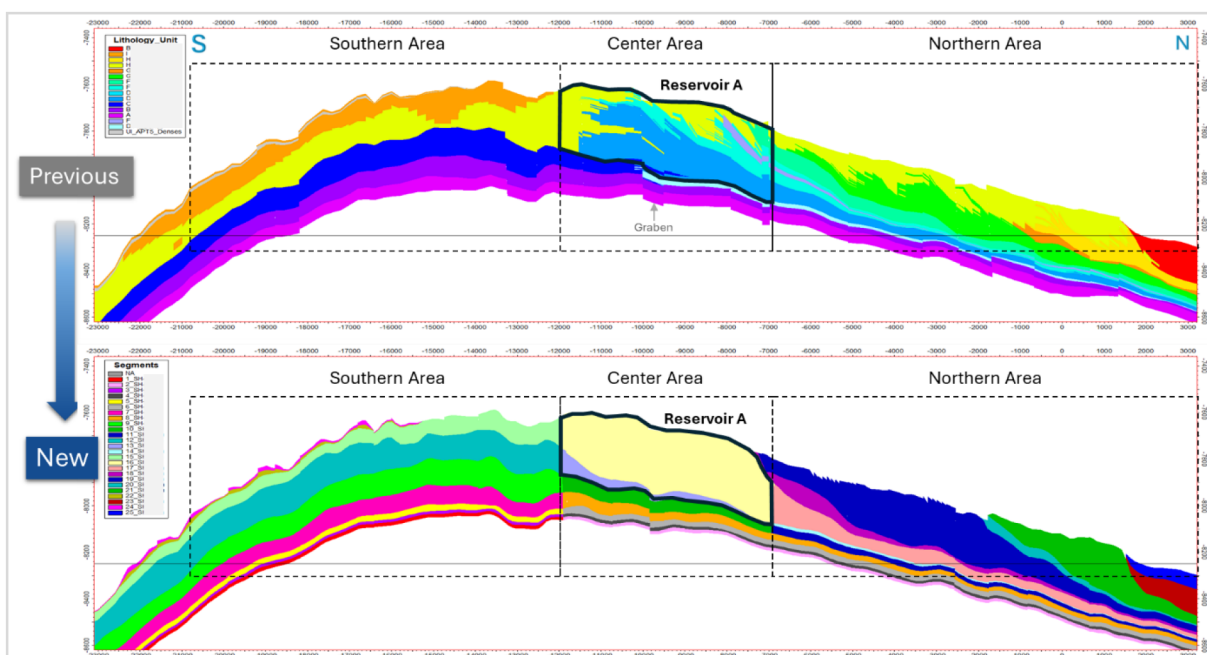
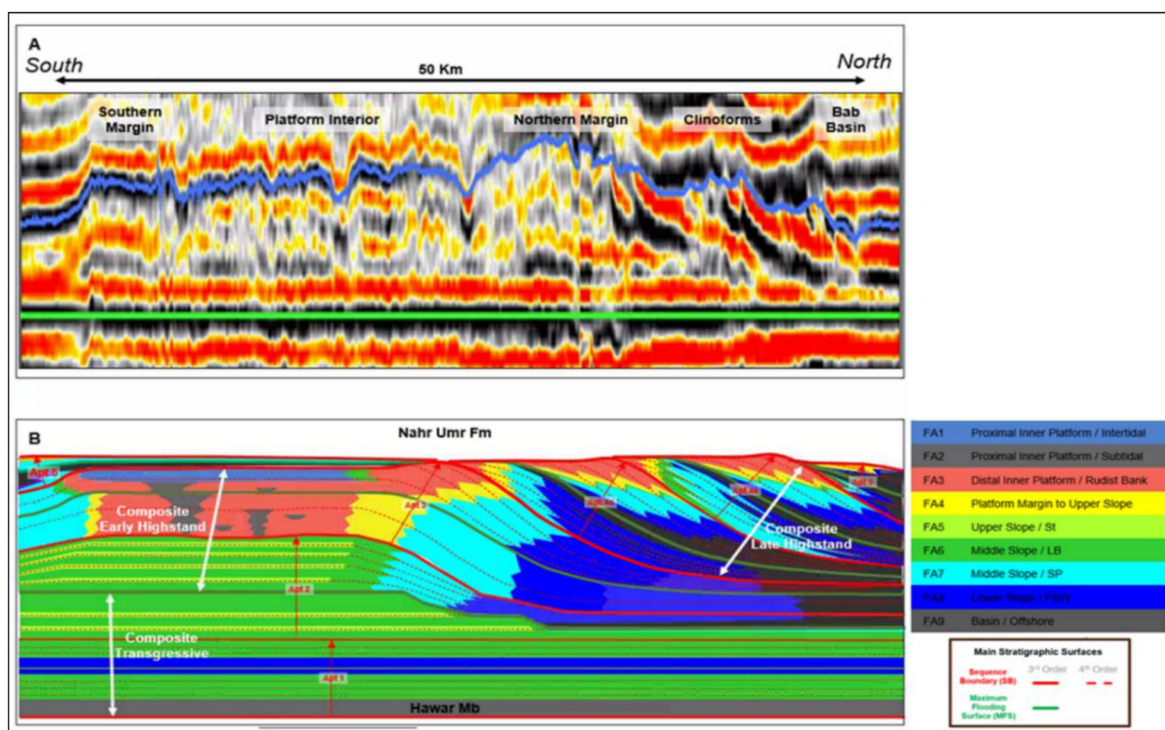
Shuaiba Formation is a major regional reservoir with significant heterogeneity in facies and pore systems. This complexity results in poor areal and vertical sweep when water injection is applied, contributing to pressure decline and increasing water cut. Reservoir A in Shuaiba formation, described in this paper, contributes around 35% of field production and exemplifies these challenges. Pressure decline began in 2017 following a 20% reduction in injection and worsened after 2019, with a further increase of 10% in production. Attempts to raise injection rates delivered only partial improvement, confirming the need for a revised Field Development plan (FDP).

This paper presents how a revised FDP, supported by enhanced surveillance and automated monitoring, was implemented to reverse pressure decline, manage rising water cut, and sustain production in this mature Shuaiba reservoir.

Geological Setting

The Shuaiba Formation, part of the Lower Cretaceous Thamama Group, is a regionally extensive carbonate unit deposited during the Early Aptian across the Arabian Plate. In the study area, it overlies the dense limestones of the Kharia Formation (Hawar Member) and is unconformably overlain by the Nahr Umr Formation, marking the Aptian- Albian boundary. This contact reflects a major shift from carbonate to siliciclastic deposition, associated with regional sea-level fall and tectonic reorganization.

Originally interpreted within a lithostratigraphic framework, the Shuaiba Formation has been redefined using a high- resolution sequence stratigraphic approach (Fig. 1). This shift enabled the identification of elementary sequences and key stratigraphic surfaces—such as maximum flooding surfaces and correlative conformities—which revealed the presence of vertical flow barriers and improved the delineation of laterally continuous flow zones (Fig. 2). The refined framework enhanced the correlation of depositional environments and provided a more accurate prediction of reservoir heterogeneity and connectivity. Furthermore, integration with Formation pressure test data (MDT) and production data confirmed the existence of stratigraphic layers acting as partial or complete baffles to flow, offering a clearer understanding of reservoir compartmentalization and dynamic behavior.



Deposited along a carbonate ramp system, the Shuaiba Formation exhibits a tripartite facies distribution: inner ramp (rudist banks and microbial carbonates), mid ramp (bioclastic and peloidal sediments), and outer ramp (fine-grained limestones and shales). These environments were strongly influenced by eustatic sea-level fluctuations, which controlled accommodation space and sedimentary energy regimes. Highstand

conditions favored rudist-dominated facies with excellent reservoir quality, while transgressive phases led to the deposition of finer, lower-energy facies with limited reservoir potential.

The formation is geographically divided into southern, central, and northern sectors. The southern region is dominated by high-energy rudist facies, forming the main reservoir units. The central region represents a transitional zone with mixed facies and moderate reservoir quality, while the northern sector is characterized by deeper marine facies, including the dense Bab Member, acting as a non-reservoir interval. Reservoir A is located in the central area.

This stratigraphic reinterpretation, supported by core, log, and seismic data, has significantly enhanced the understanding of the Shuaiba Formation's internal architecture. It provides a robust basis for reservoir zonation, flow unit definition, and development planning in one of the region's most prolific carbonate systems.

Methodology

Data validation and Automation

The first step was to ensure reliability and consistency in the reservoir surveillance dataset. Wells equipped with permanent downhole gauges (PDHGs) provided high-frequency pressure and temperature data, while non-PDHG wells were monitored through periodic surveys (BHFP, BHCIP, PBU). All data is centralized in the Prosource system.

In this phase, the reservoir leveraged the outcomes of the Digitalization Pilot of PDHG with Kappa Automation (Martinez et al., 2025), which introduced automated pressure transient analysis (PTA) using PDHG data (Fig. 3). This solution reduced manual interpretation, enabled automatic shut-in detection, and provided faster results (e.g., pressure, permeability, skin). This automation improved the efficiency of surveillance workflows and directly supported the integration of reliable pressure data into the dashboards used in the study.

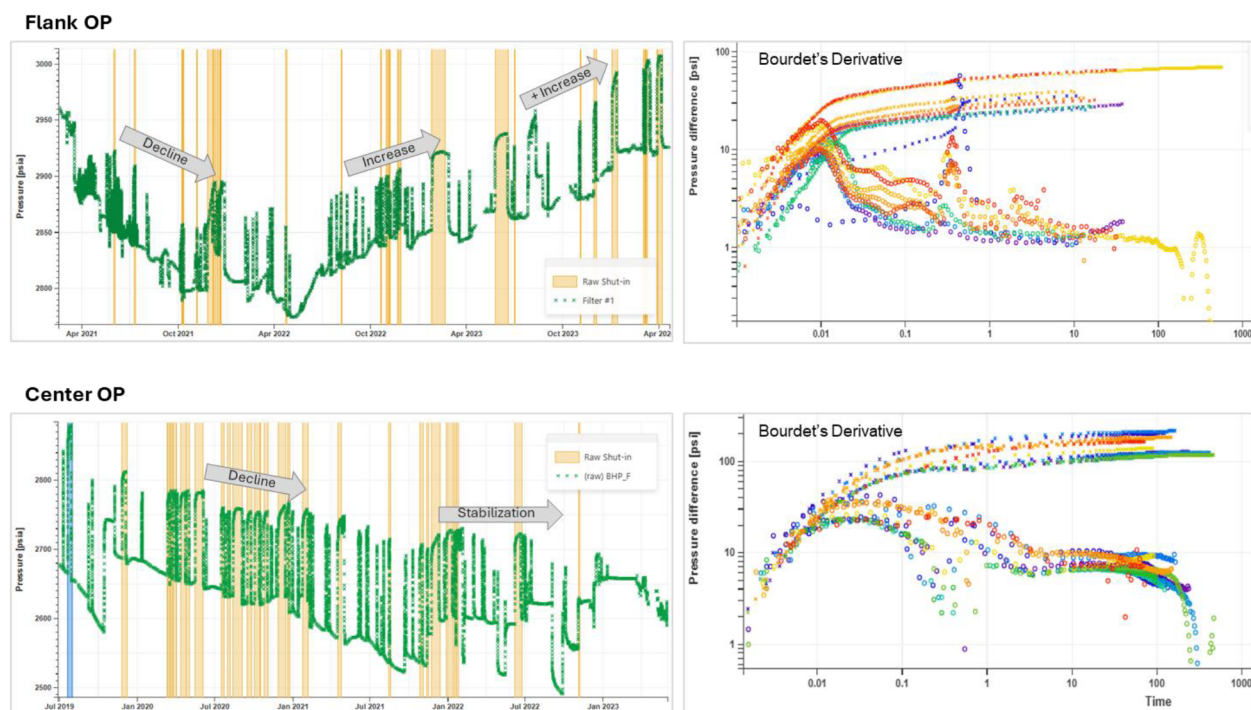


Figure 3—Examples of reservoir pressure plots from Kappa automation displaying PDHG data with identified shut-in's and Bourdet's derivative. The upper plot shows good pressure restoration in a flank producer, while pressure stabilization is reported in the lower plot corresponding to a well located at the center area.

Currently, 15% out of the total producers have PDHG deployed, and this number is expected to increase as part of the ongoing automation program in the field.

For the non-PDHG wells, the workflow was optimized to maximize data quality while minimizing production deferment:

- **Shut-in duration** for static pressure was redesigned. Analysis of PBU data from previous campaigns showed that stabilization required 60% less shut-in duration. This optimization reduced unnecessary downtime, enabled more frequent surveys, and increased data acquisition in more wells.
- **Data loading** of pressure surveys results was optimized through a coordinated task between Surveillance and IT Team, ensuring that validated results were available in the database within one week of acquisition.
- **Reporting templates** for pressure survey results were reviewed and updated to ensure all relevant information was captured and properly documented.

Integration into Dashboards

Following the data validation and automation stage, the surveillance dataset was integrated into an interactive visualization platform. A series of Spotfire dashboards were developed to analyze pressure behavior across multiple scales—well, sector, and full reservoir level.

The Spotfire project was directly linked to the Prosource database to access validated surveillance and production data (Fig. 4). In parallel, a master Excel file was generated and linked to the Spotfire project to compile additional tags for flexible filtering, allowing new variables to be added for analysis.

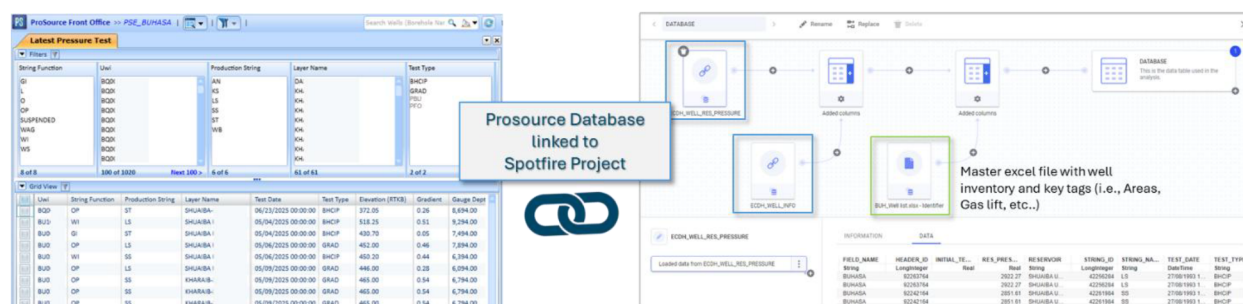


Figure 4—Direct connection of Prosource data source and Spotfire project. The link refreshes automatically with each data update. Master excel file is generated to add custom well tags enabling flexible filtering and detailed analysis.

- Well level, time-series plots were configured to track pressure evolution for each string (Fig. 5).
- Sector level, dashboards provided yearly and monthly pressure profiles, for side-by-side comparison between reservoir sectors and sub-sectors/areas (e.g., east, west, center, flank in Fig. 7).
- Reservoir level, isobaric and variation-rate maps provided an integrated view of pressure distribution and decline rates across the field (Fig. 10). Isobaric maps are generated from Spotfire using Python code in JupyterHub then loaded in Spotfire.

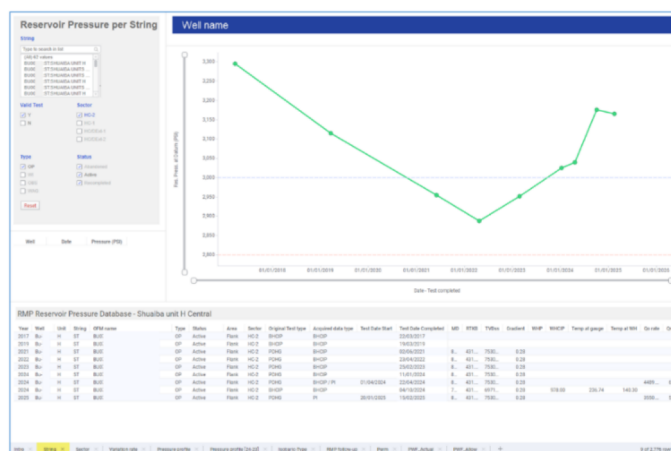


Figure 5—Spotfire dashboard for reservoir pressure data per string. Pressure trends are displayed and detailed information at the bottom. Filters were created using html code, allowing flexible selection.

Spotfire was selected for its flexibility and advanced visualization capabilities compared to conventional platforms. Dashboards provide instant updates once data is loaded, requiring minimal manual manipulation, and supporting interactive workflows through filtering. These features ensure consistent and efficient use of surveillance data, while extending access across disciplines to enable collaborative decision-making.

Subsurface review

A comprehensive subsurface overview was carried out to reassess reservoir performance and support an updated FDP. The workflow included:

- FDP reassessment: Initial review of the Long-term development strategy, triggered by pressure decline and rising water cut.
- Dynamic Sectorization: Divide the reservoir into sectors by integrating geological features with dynamic responses such as pressure communication and flow behavior.
- Numerical Simulation: Updated simulation model was used to optimize water injection strategy, evaluate development scenarios (e.g., roll up and line drive patterns) and support sectorization analysis.
- Assessment of underperforming injectors: Identified low-efficiency injectors for stimulation.
- Voidage Replacement Ratio (VRR) management: Production and injection balancing along flow units within Reservoir A.

All the insights were consolidated into the updated FDP, ensuring a robust strategy for restoring pressure and sustaining production.

Surface review

A major surface review was initiated in 2019 to evaluate field water injection capacity and identify bottlenecks that limit Shuaiba's reservoirs waterflood performance, including Reservoir A. The assessment covered the entire injection system—from supply wells to pumps, clusters, and well injectors—comparing actual capacity to current and future requirements.

The methodology followed a two-stage approach:

- Stage 1 [2020- Q1/2024] - Major capacity restoration and expansion
Focused on closing large capacity gaps by bringing additional water volumes on stream through new supply wells, facilities upgrades, and cluster expansion. The objective was to restore injection

to the required level and prepare the system for ramp-up injection, especially in Reservoir A (Roll-up strategy).

- Stage 2 [Q3/2024-2025] - Capacity alignment and optimization

Triggered by prolonged pressure recovery and unplanned production growth in 2025-2026, which increased injection demand beyond the original plan. Stage 2 addressed residual capacity gaps arising from accelerated injection and higher offtake. Actions focused on water re-balancing across clusters and reservoirs, cluster connectivity improvements, and quick-win measures to reinforce system resilience and ensure water supply kept pace with updated requirements.

Results

Dynamic Sectorization

Unlike static sectorization (based on rock properties, faults, or facies), dynamic sectorization adapts to how the reservoir responds during production and injection.

The new geological framework directly supported the new dynamic sectorization (Fig. 6).

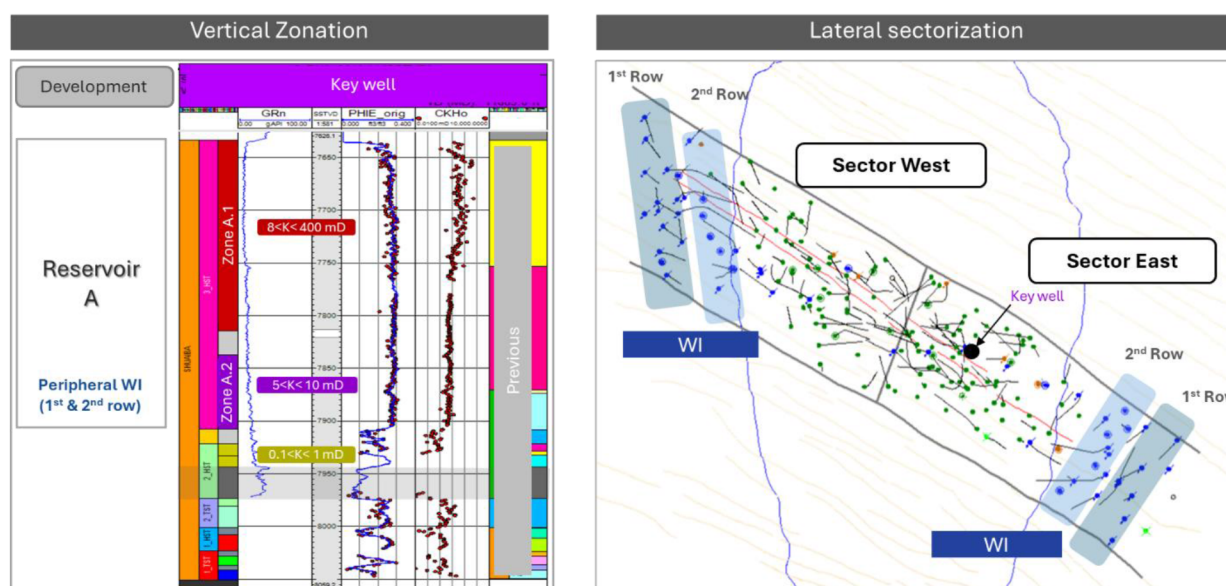


Figure 6—New dynamic sectorization implemented in Reservoir A with 2 vertical zones and 2 lateral sectors. A key well type is also presented, integrating sequence stratigraphy, flow zones, and core data (porosity, permeability).

- Vertical zonation: In Reservoir A, two flow units, A.1 and A.2, were identified. They are vertically communicated, with no major barriers and confirmed by MDT data. Additionally, towards to the north, the reservoir underlying Reservoir A transition into a dense zone with poorer porosity and permeability.
- Lateral sectorization: Two sectors were defined - East and West. This division reflects the symmetry of reservoir development (peripheral water injection). Both sectors show similar distribution of properties from South to North, well conformance, pressure performance from the flanks towards the center and water advancement that progresses comparably towards the pool.

This sectorization allowed better reservoir management, improved surveillance and monitoring, and provided support for history matching and model calibration, an approach that is consistent with recommendations from other mature carbonate waterflood projects in the region.

Reservoir pressure analysis

Analysis of reservoir pressure data was performed using Spotfire dashboards built for Reservoir A. Plots were displayed honoring new sectorization and producers were sub-divided into flank and center areas, so trends could be easily detected (Fig. 7).

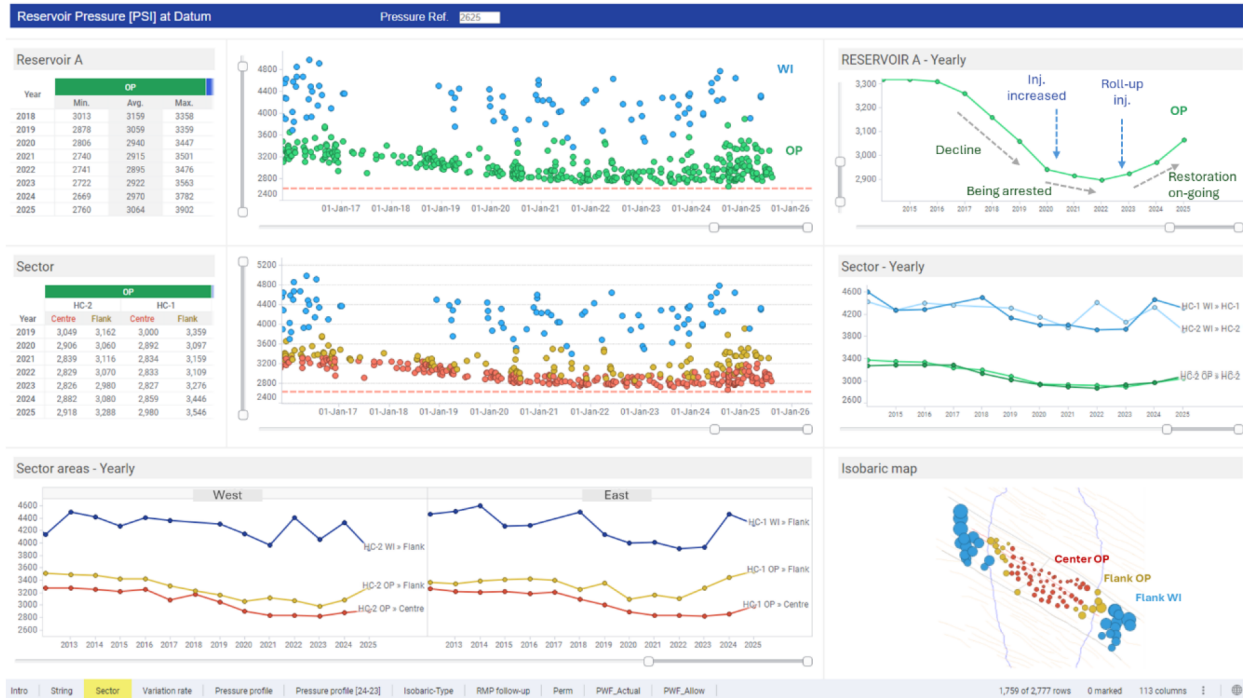


Figure 7—Spotfire dashboard showing Reservoir A across sectors East and West, and areas such as flank and center.

Key observations from the dashboard:

- Reservoir level: Pressure trend changed in 2021 following an injection rate increased in flow unit A.1 through existing wells, however, was insufficient. The revised FDP implemented the roll-up injection with dual injectors targeting both flow units, A.1 and A.2. The strategy proved to be efficient, increasing 140 PSI by 2025 since its implementation.
- Sector level: East and West shows similar values and restoring trends as a result of close VRR management.
- Flank vs. center: Center wells in both sectors showed a similar trend of stabilization with gradual improvement, which initiated the second integrated review in 2024 to accelerate the pressure recovery (anticipating higher production targets in 2025). In flank producers, sector East recovered faster than ones in sector West, primarily linked to the earlier commissioning of roll-up injectors in the East and less injection capacity constraints.

The variation rate dashboard was helpful in quantifying the decline or increasing rate. This can be defined as the difference between the last pressure and previous pressure divided by number of months, expressed in PSI/month.

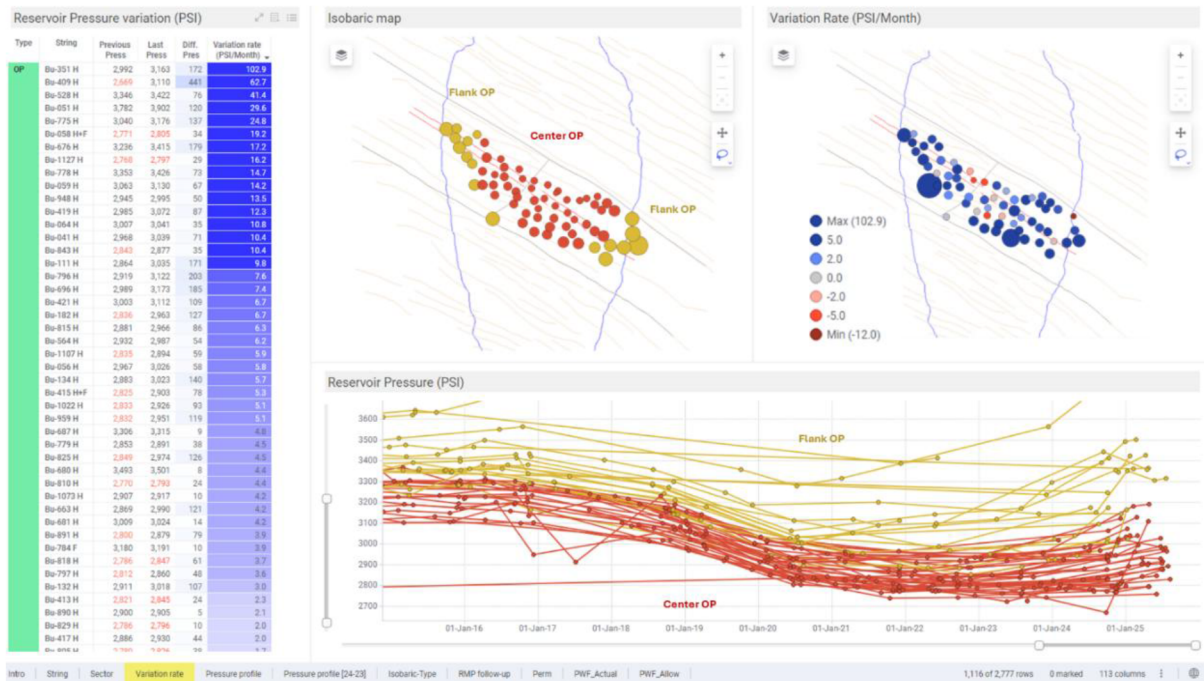


Figure 8—Spotfire dashboard of variation rate in Producers (Difference between last pressure and previous pressure divided by number of months) expressed in PSI/Month to quickly identify any decline or increasing trend.

On the other hand, reservoir pressure was not monitored as planned at that time due to high demand production that prevents the wells from closing for static pressure acquisition. As mentioned in the Methodology section, the shut-in period was optimized and allowed more frequent surveys. RMP activity in 2024 increased in producers (Fig. 9) to closely monitor pressure performance, and activity in 2025 is expected to surpass 2024.

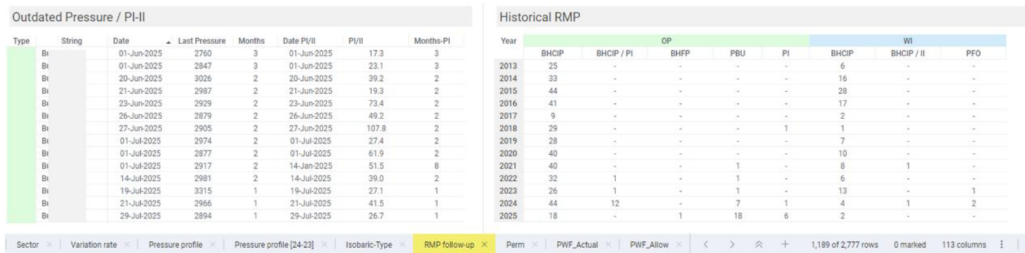


Figure 9—Spotfire dashboard to follow up Reservoir Monitoring Program (RMP)

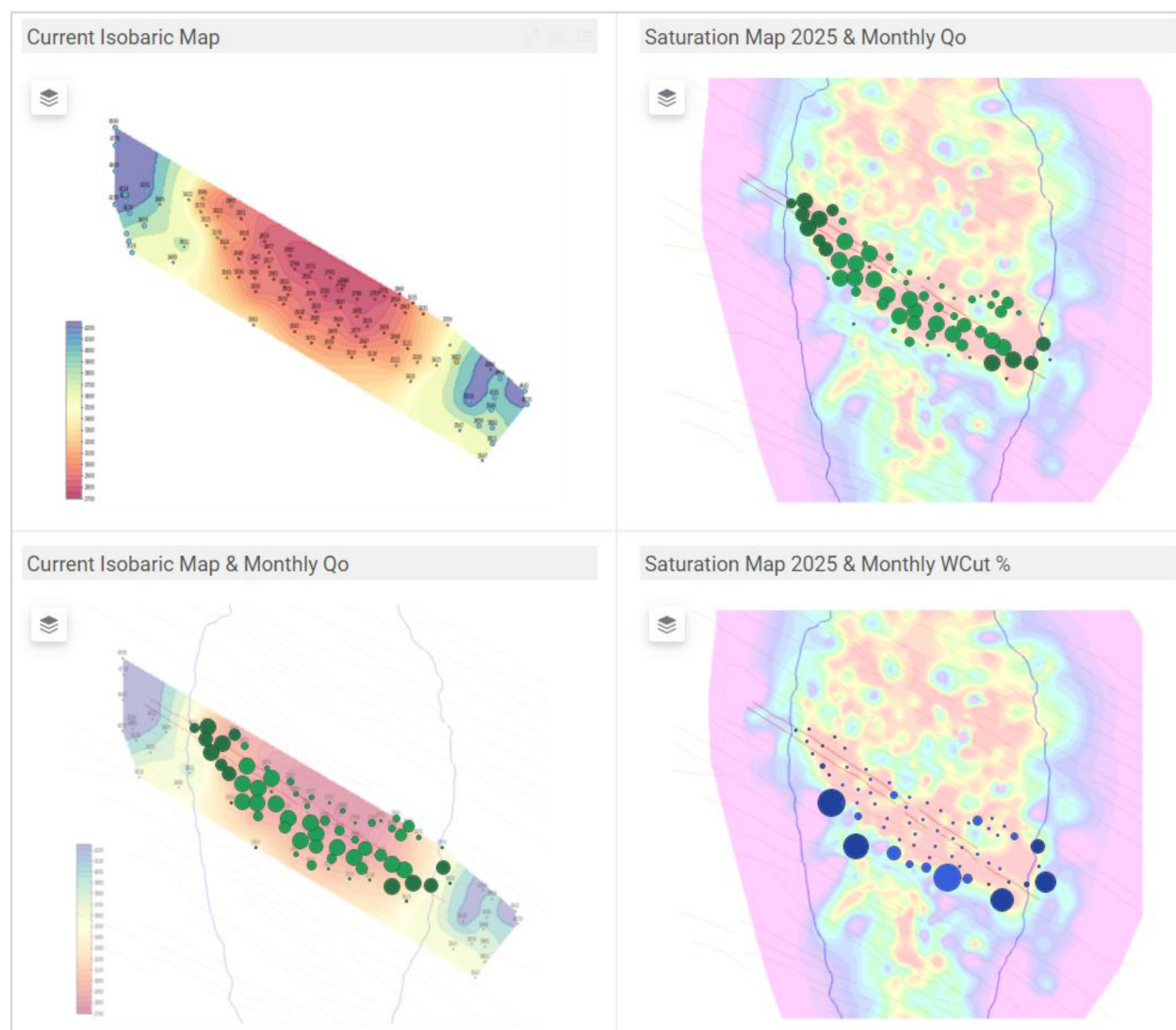


Figure 10—Spotfire display. Left: Isobaric maps displaying Monthly oil rate. Right: Saturation maps displaying Oil rate and Water cut

Water Injection Capacity and Debottlenecking

The major surface review delivered significant improvements in water injection capacity across the field that overcame the reduction in injection experienced in 2017 and 2018 (Fig. 11).

- Stage 1 [2020- Q1/2024] - Major capacity restoration and expansion
 - This stage delivered a step-change in injection performance by combining new water supply, infrastructure upgrades, and system reliability improvements:
 - Water supply expansion: Drilling of new Water source wells (WSWs), replacement of aged WSWs, and commissioning of 10 new clusters adding ~175 MBWPD.
 - Efficiency upgrade: Installation of Seven Variable Speed Drive (VSD) adding ~20 MBWPD and enhancing operational flexibility.
 - Connectivity enhancements: Completion of three transfer lines South - North to improve inter-cluster water distribution and balance injection across the field.
 - Well injectors: WO to restore integrity and reactivate injectors, Stimulation campaign to restore injectivity and new WI aligned with revised FDP.

- Stage 2 [Q3/2024-2025] - Capacity alignment and optimization

Stage 2 was initiated as pressure restoration in key reservoirs (Reservoir A as main contributor) was taking longer than expected, while production growth in 2025 - 2026 created additional injection demand that had not been anticipated in the original plan. This stage complemented stage 1 by targeting residual bottlenecks, closing emerging gaps and fine-tuning distribution:

- Quick capacity gains: Six water injection wells were fast-tracked in the central area, including four in Reservoir A (~20 MBWPD).
- Debottlenecking through supply well reactivation, pump upgrades, and cluster interconnections adding ~75 MBWPD.
- Sweep efficiency: Injection was optimized laterally and vertically to improve volumetric support based on dynamic model results: Injection from the peripheral scheme was reduced and re-allocated to roll-up injectors, and vertically, injection from A.1 was reduced and re-allocated to increase A.2 (lower permeability) that will delay water breakthrough.
- Well injector performance: Compliance analysis of 400+ injectors (full field) identified two main constraints, Table 1: Back-pressure, and cluster capacity. A dedicated stimulation campaign targeted injectors with back pressure issues, achieving 65% increase in total injection rate with 90% success in Reservoir A.

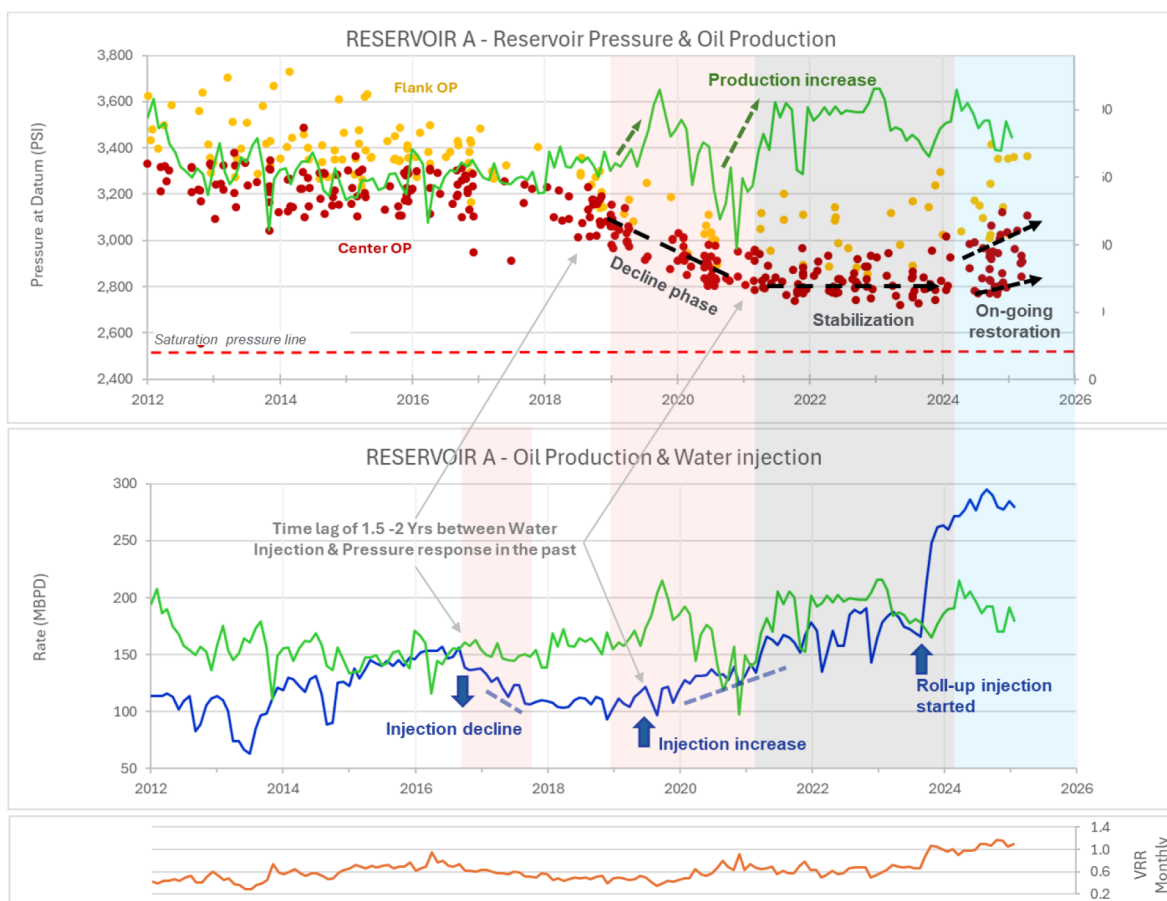


Figure 11—Last 13 years of historical performance of Reservoir A, depicting oil production and water injection along reservoir pressure highlighting main events.

Table 1—Stage 2: Well injector - Compliance summary and Actions

Well Compliance	Description	Actions
Full compliance	Injection as per requirement	No Action
Above requirement	Injection above target requirement	Flowrate to be adjusted as per requirement
Below requirement (Back pressure)	Well cannot inject at full flow due to back pressure constraints	Stimulation campaign in Reservoir A increased 65% in total injection rate for treated wells.
Below requirement (Cluster capacity)	Well cannot inject at full flow due to insufficient water availability from the supply cluster	Debottlenecking through reactivation of supply wells, pump upgrades, and cluster interconnections added ~75 MBWPD.
Closed temporary	Not injecting for less than 1 year	Mostly due to HSE policy during Drilling activities, or under reservoir monitoring program. Action planned to expedite reactivation.
Closed permanently	Not injecting for 1 year or more	Integrity assessment conducted. Wells either to be reactivated, utilized in another reservoir or decommissioned as required

In the medium-term, an ongoing project will add significant treated seawater volumes, further strengthening injection capacity.

Revised Field Development Plan

Carbonate reservoirs in the Middle East face strong heterogeneity, complex fracture systems, and variable waterflood efficiency, making adaptive development a necessity rather than an option. For the Shuaiba reservoir A, this philosophy was applied by incorporating new geological interpretations (flow units), sectorization (combining a static and dynamic approach), and smart pressure surveillance to refine the Field Development Plan (FDP) and optimize water injection strategies.

Reservoir A is a mature carbonate reservoir under peripheral water injection, developed with slanted producers and injectors in its majority. Spacing ranges from 550m. to 750m. between producers. Reservoir rock properties show good porosity ranging between 22% to 28% along reservoir A, and permeability ranging from 8mD to 400mD for A.1 and for A.2 from 5mD to 10mD (Fig. 6).

The field experienced a reduction in water injection in 2017 due to surface capacity issues, and reservoir pressure in Reservoir A started to response after ~1.0 year with a steady decline. By 2019, the team embarked in an integrated surface-subsurface review to reassess injection strategy and redesign the FDP. The objective was to realign water injection capacity with production growth while ensuring surface facilities could accommodate water requirements.

As described in Stage 1 of Water injection capacity and debottlenecking program, major actions were executed, including new WSW, reactivation of injectors, improved injectivity through stimulations and facilities upgrades. By 2021, reservoir pressure responded positively after ~1.5 years since an increase in injection started, showing clear stabilization, despite ongoing high offtake. Continuing with implementation of revised FDP, 13 new dual water injectors (A.1, A.2) were drilled and commissioned by late 2023. All this effort managed to increase the total water injection rate by 155% (from 2019 to 2024). These measures successfully reversed the decline, stabilized reservoir pressure, initiated ongoing pressure restoration, and provided the surface capacity needed to sustain roll-up injection (Fig. 11).

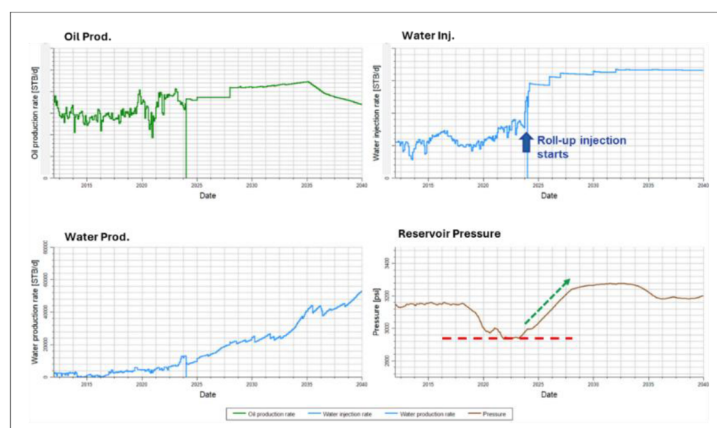


Figure 12—Simulation results of roll-up injection strategy in Reservoir A

Well level dashboard in Fig. 13. displays producers honoring their geographical location in a West - East cross section. The comparison between the 2024 and 2023 pressure values shows an overall increase, with greater recovery in the East due to earlier and higher injection support (fewer capacity constraints). The improvement aligns with the start of roll-up injection in 2024, confirming the efficiency of this strategy. In contrast, negative variation rates were observed in a few wells, indicating localized pressure decline mainly driven by high offtake.

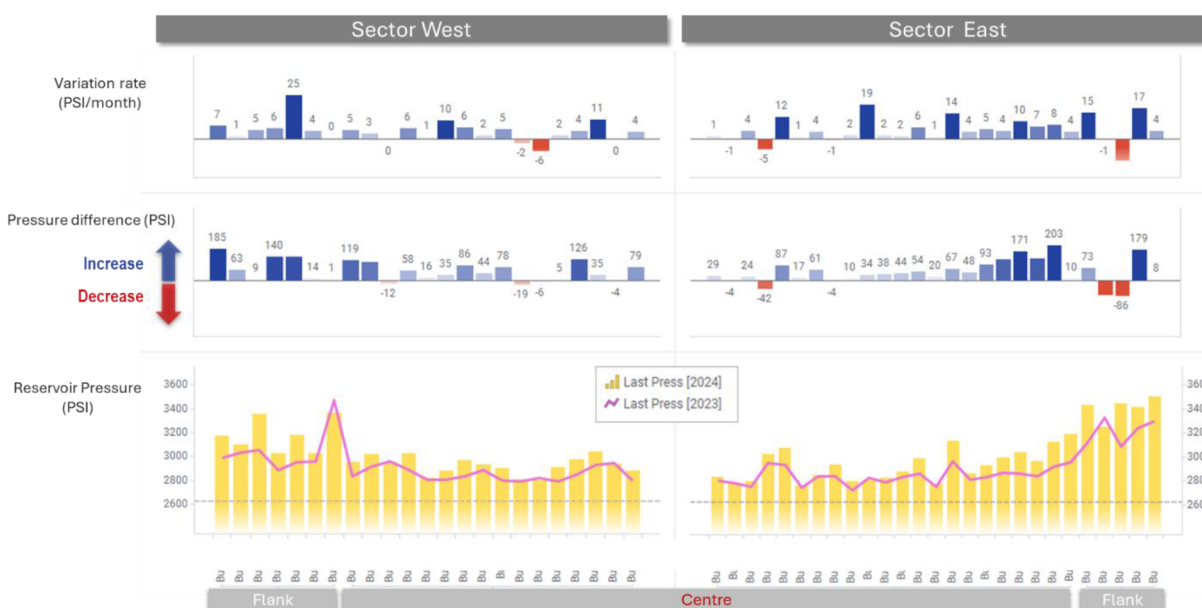


Figure 13—Spotfire dashboard of Reservoir pressure - Oil producers in a cross-section comparing 2024 vs. 2023.

These results confirm the effectiveness of the revised development plan and the importance of timely production and injection balancing.

However, a second wave of production growth was accelerated to 2025 - 2026, creating new reservoir pressure challenges. In 2024, another integrated review was conducted to anticipate these requirements and ensure timely actions for production target compliance and pressure restoration, especially at the center area. Based on numerical simulation, four new horizontal water injectors, part of the long-term FDP, were optimized and accelerated, targeting the lower part of A.2, all WI at the center with an average horizontal exposure of 2500 ft (Fig. 14). Their objective was to provide pressure support from the bottom of A.2 to

A.1. This configuration enhances pressure support in A1 (main contributor); delays water breakthrough and ensures more efficient sweeping in the main producing zone.

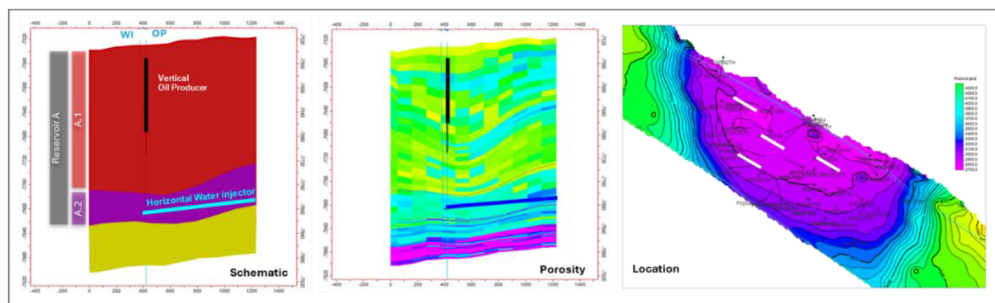


Figure 14—Left: New WI schematic configuration targeting bottom of A.2. Right: Location of 4 New WI t with isobaric map in the background

In parallel, outcomes from the Stage 2 of Water injection capacity and debottlenecking program secured additional required injection by adding 75 MBWPD capacity and reallocating injection to enhance sweep efficiency. A compliance analysis of injectors resolved back-pressure issues, achieving a 65% rate increase through a dedicated stimulation campaign.

Fig. 15 shows the ongoing pressure restoration, with positive increases already observed in center producers following the commissioning of 4 new injectors in 2025. This dashboard allows rapid correlation of well performance with rate changes and ensures producers comply with internal guidelines to maintain bottom-hole (BHFP) above saturation pressure plus 100 psi. For now, water cut remains stable and does not represent a risk, however, gas lift activation is planned for production continuity.

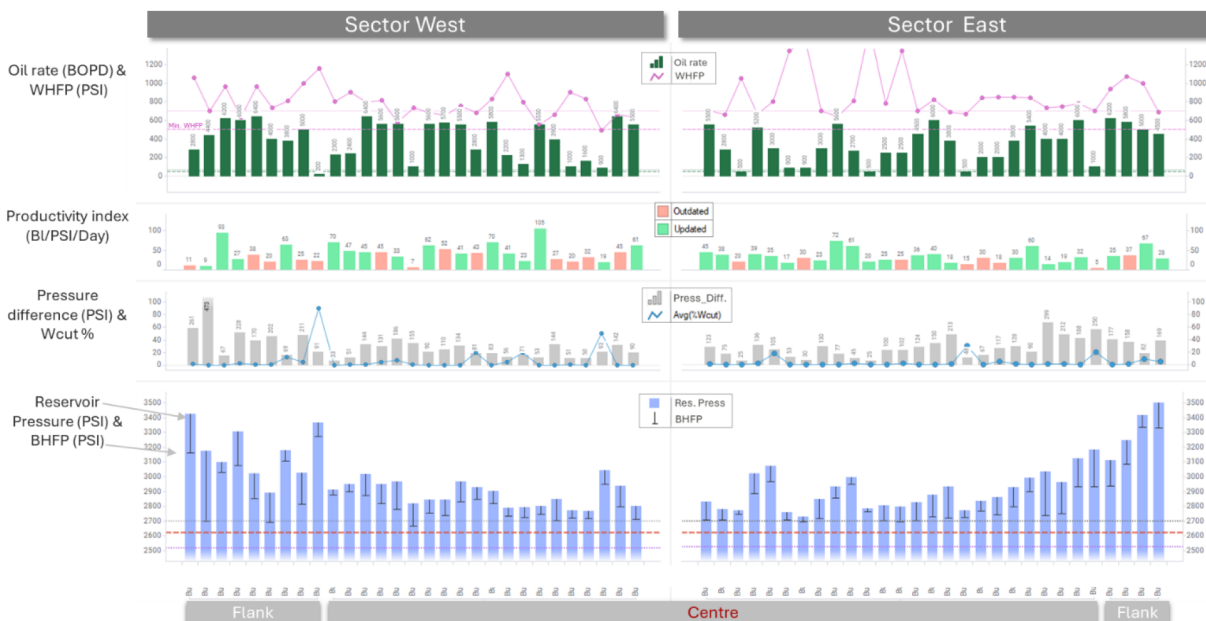


Figure 15—Spotfire dashboard showing Reservoir pressure and BHFP linked to oil production, Wcut and PI, allowing fast and close monitoring.

Conclusion

The implementation of an adaptive field development strategy in Reservoir A of the Shuaiba Formation has proven effective in reversing pressure decline and sustaining production under high-demand conditions. By

integrating updated geological interpretations, dynamic sectorization, and smart surveillance workflows, the team was able to realign water injection with reservoir needs and surface capacity.

Adaptive Field development Strategy

- Reservoir pressure decline was successfully reversed through a revised field development plan that integrated updated geological understanding, dynamic sectorization, optimized injection distribution, and smart surveillance.
- Roll-up injection scheme with dual WI targeting A.1 and A.2 flow units, stabilized and increased reservoir pressure by 140 PSI in average from 2020 to 2024.
- Dynamic sectorization improved reservoir management: Vertical zonation and lateral sectorization on Reservoir A helped optimize injection and reservoir monitoring.
- Revised FDP aligned injection requirements with production growth, ensuring pressure recovery and reservoir health.
- Introduction of new line-drive horizontal injectors targeting the lower part of A.2 strengthened pressure support in the center area and it is expected to delay water breakthrough.
- Surface upgrades closed water distribution gaps: Cluster optimization and interconnections improved water availability.

Smart Surveillance and Monitoring

- PDHG data and Spotfire dashboards supported rapid decision-making and efficient cross-disciplinary collaboration: Pressure trends, variation rates, and RMP activity were tracked across well, sector, and reservoir levels.
- Shut-in time for static pressure survey was optimized by 50%, allowing more frequent pressure surveys and facilitate the monitoring especially after periods of high demand.

Injection Surface Capacity

- Injection rate ramped up by 155% supported by increase in water capacity, aligning with injection requirements.
- New clusters, stimulation campaigns, surface debottlenecking and water supply expansion, collectively contributed to this substantial capacity increase.
- Compliance analysis identified and resolved injectivity issues.

This integrated strategy proved effective in sustaining production under high-rate conditions, validating the revised FDP and ensuring reservoir health.

Acknowledgements

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Nomenclature

FDP-	Field Development Plan
VRR-	Voidage Replacement Ratio
RMP-	Reservoir Monitoring Program
PDHG-	Permanent Downhole Gauge
BHFP-	Bottomhole Flowing Pressure

BHCIP- Bottomhole Closed-In Pressure
PBU- Pressure Build-Up
PTA- Pressure Transient Analysis
WSW- Water Source Well
VSD- Variable Speed Drive
OP- Oil producer , WI - Water Injector
HSE- Health, Safety, and Environment
MBWPD- Thousand Barrels of Water Per Day
PI- Productivity Index

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